

Arsha Niksa

✉ a.niksa@sharif.edu 🌐 aniksa.me in arsha-niksa 📷 A-Niksa

Education

Sharif University of Technology

Oct 2021 – Jul 2026

BSc (Double Major), Mathematics and Physics

- GPA: 19.54/20.00 (Rank 1 in Mathematics)

Coursework

Graduate

- **Mathematics:** Algebraic Geometry (17.00/20.00), Topics in Mathematical Biology (19.00/20.00), Algebraic Topology (19.00/20.00), Commutative Algebra (Audited)

Undergraduate

- **Computer Science:** Data Structures (20.00/20.00), Advanced Programming (20.00/20.00)
- **Mathematics:** Statistics (20.00/20.00), Complex Analysis (20.00/20.00), Elementary Diff Geometry (19.40/20.00), Topology (20.00/20.00), Abstract Algebra (18.80/20.00), Linear Algebra (20.00/20.00)
- **Physics:** Thermodynamics (20.00/20.00), Electromagnetism (20.00/20.00), Group Theory in Physics (20.00/20.00)

Awards

Bronze Medal (Rank 3), National University Mathematics Olympiad 2024

Top 0.2%, National University Entrance Exam (STEM Stream) 2021

Gold Medal, Persian Young Physicists' Tournament 2020

Gold Medal, Korea-USA Science and Engineering Federation 2020

Gold Medal, Korea Science and Engineering Fair 2019

Academic Experiences

Research Project — Manifold Reconstruction from Morse Critical Points

Remote

With Vidit Nanda (University of Oxford)

Jul 2026 – Present

- Studying reconstruction of embedded manifolds from Morse critical-point information of directional height functions, including conditions under which dense critical data determines the manifold.
- Developing computational experiments that reconstruct from point-cloud-derived critical data and evaluate Hausdorff error between the original and reconstructed manifolds.

Research Project — Symmetry-Invariant Kernels for Molecular Shape Spaces

Remote

With Marina Meilă (University of Waterloo and Vector Institute)

Mar 2026 – Present

- Investigating kernels on molecular shape spaces that are invariant under symmetry-group actions, with applications to molecular dynamics data.
- Developing representations that compare molecular configurations while factoring out symmetry-equivalent transformations.

Research Project — Manifold Cohomology Estimation

Remote

With J. Jost and M. Eidi (Max Planck Institute)

Jul 2025 – Feb 2026

- Developed methods for estimating manifold cohomology from sampled point clouds, extending ideas from data-driven estimation of real cohomology groups.
- Explored dimension-reduction methods designed to preserve geometric and topological structure.

Directed Reading — Topological Data Analysis

Remote

Under H. Edelsbrunner (Institute of Science and Technology Austria)

Jul 2024 – Mar 2025

- Studied computational topology, stability theorems in persistent homology, persistence modules, and algebraic stability.

Directed Reading/Research — Geometric Formalisms of Mechanics

Remote

Under T. Strobl and H. Nahari (Camille Jordan Institute)

Feb 2024 – Aug 2024

- Studied singular Riemannian foliations, I-Poisson manifolds, Dirac structures, and advanced differential geometry.

Professional Experience

Data Scientist — Searchwise

Tehran, Iran

Full-time, On-site

Jul 2026 – Present

Contributing to the development and evaluation of data-driven and language-related systems, with a focus on benchmarking, data pipelines, and deployment.

- Worked on the end-to-end development of SnappPay's spell-checking system.
- Built and maintained data pipelines used across experimentation and evaluation workflows.
- Dockerized services and supported deployment and production workflows.
- Contributed to DevOps-related tasks, including service configuration and environment management.

Teaching Experiences

Teaching High-school Students: Advanced Physics for International Young Physicists' Tournament preparation.

Teaching English as a Foreign Language: English instructor for children and teenagers at Hermes Language Institute.

Publications

Persistent Patterns in Eye Movements: A Topological Approach to Emotion Recognition

Jul 2025

Arsha Niksa, Hooman Zare, Ali Shahrabi, Hanieh Hatami, Mohammadreza Razvan

[arXiv Preprint](#) [↗](#)

Effects of Roughness on Rarefied Gas Flow in a Microfluidic Channel

Sep 2021

Arsha Niksa, Amirreza Soheili, Vahid Bazargan

[Proceedings of the 19th Fluid Dyn. Conf., SUT](#) [↗](#)

Test Scores

- GRE Mathematics Subject Test: **850**, 82nd Percentile
- iBT TOEFL: **119/120** (Reading: 30, Listening: 30, Speaking: 29, Writing: 30)

Skills

Programming (In Order of Proficiency): Java, Python, R, Julia, MATLAB, C++

Soft: Public Speaking, Team Collaboration, Leadership

Languages: Persian (Native), Kurdish (Native), English (Fluent), German (Beginner), French (Beginner)

Volunteering

Staff Member of the Women in Mathematics Seminars

Apr 2024

- Hamband, the Mathematical Association of Sharif University, coordinated a number of talks highlighting the role of women in Mathematics.
- Was a staff member, helping with executive tasks.

Student Resources Coordinator at CaYPT

Sep 2021, Sep 2023

- Managed and advanced the computational tutorials project at the CaYPT Committee (the Canadian Young Physicists' Tournament) as a member of STEM Fellowship, a Canadian non-profit.
- Prepared materials and video tutorials, published on the YouTube channel of the tournament.

External Science Communications Team Lead at STEM Fellowship

Nov 2020, Oct 2021

- Led the team working on securing sponsorships from Canadian corporations and building partnerships between STEM Fellowship and other North American non-profits for STEM outreach programs.
- Focused on programs for underrepresented communities and the Indigenous communities of Canada.

Hobbies

Ping Pong, Guitar